A PROJECT TITLE

ON

Fake Product Review Detection Using NLP and Random Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

Name:A.Lakshmi Sai Teja **roll:**2520080005

Name:D.Vignesh Kumar **roll:**2520090071

Name:P.Vedadri **roll:**2520090060

Under the Esteemed guidance of

Dr.Y.LakshmiPrasanna

Assistant Professor, CSIT

<CSIT>

KLH UNIVERSITY (DEEMED TO BE UNIVERSITY)

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

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DEPARTMENT OF Computer Science and Information Technology.

Name of Title: Fake Product Review Detection Using NLP and Random Forest

1. Abstract:

The exponential growth of e-commerce has made customer reviews a foundational element of consumer trust and purchasing behavior. However, this ecosystem is increasingly threatened by fake reviews generated by malicious actors and automated bots. This project presents a comprehensive, end-to-end machine learning system designed to detect fraudulent product reviews using Natural Language Processing (NLP) and a Random Forest classifier.

The system architecture follows a rigorous lifecycle pipeline. Unstructured review text undergoes systematic NLP preprocessing—including lowercasing, stop-word removal, and lemmatization—before being transformed into numerical representations using TF-IDF vectorization. These statistical vectors are enriched with engineered linguistic meta-features, such as sentiment polarity and stylometric markers, to capture the structural "tells" of deceptive writing.

To establish a performance baseline and analyze the bias-variance trade-off, a Logistic Regression model with L1 and L2 regularization is initially applied. The core predictive engine then utilizes a Random Forest, which effectively manages the high-dimensional feature space and reduces ensemble variance. By extracting feature importances, the system provides algorithmic interpretability, revealing the specific linguistic patterns and vocabulary driving its predictions.

To address coordinated bot activity, unsupervised learning techniques like DBSCAN clustering and PCA are integrated to identify dense anomalies and unlabelled structural similarities among heavily templated reviews. Recognizing the severe class imbalance inherent in fraud detection, model evaluation deliberately moves beyond simple accuracy. The system is rigorously assessed using Precision, Recall, F1-Score, and PR-AUC, with hyperparameters carefully optimized via Bayesian search to minimize false positives against genuine users.

Finally, the trained model and processing pipelines are packaged and deployed as a RESTful API. The production environment includes continuous monitoring for data drift and performance degradation, ensuring the system remains resilient against evolving spam tactics and consistently safeguards platform integrity.

2. INTRODUCTION:

Online customer reviews drive e-commerce, but the rise of fake reviews severely manipulates consumer trust. This project builds an end-to-end machine learning pipeline using Natural Language Processing (NLP) and a Random Forest classifier to automatically detect fraudulent product reviews. By analyzing text vectors and deceptive linguistic "tells," this fully deployed REST API provides a scalable, highly interpretable solution to filter out malicious spam and protect marketplace integrity.

3. SOFTWARE REQUIREMENTS:

Operating System: Windows, macOS, or Linux
Language: Python 3.8+

IDE: VS Code & Jupyter Notebook

ML & Data Libraries: Scikit-Learn, Pandas, NumPy

NLP Libraries: NLTK or spaCy

API / Deployment: FastAPI or Flask

Signature of the Guide **Course Instructor**